PRATHAM CHANDRA

Senior Data Scientist | AI/ML | Generative AI | LLM Engineering | Predictive Modelling

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PROFESSIONAL SUMMARY

Senior Data Scientist with 7+ years of end-to-end experience building machine learning, Generative AI, and LLM-powered solutions across BFSI, fintech, and enterprise domains. Proven expertise in credit risk modelling, delinquency prediction, fraud detection, NLP chatbot development, and sales intelligence automation. Skilled at leading cross-functional data science teams of 6+, deploying scalable ML pipelines on PySpark and Databricks, and translating complex analytical outputs into measurable business impact. Deep hands-on experience with prompt engineering, retrieval-augmented generation (RAG), vector embeddings (FAISS), Azure OpenAI, and Format-Preserving Encryption (FPE/AES-FF3-1) for secure, production-grade AI deployment.

CORE COMPETENCIES

Machine Learning & Predictive Modelling | Credit Risk & Delinquency Prediction | Generative AI & LLM Engineering | Retrieval-Augmented Generation (RAG) | Prompt Engineering & LLM Fine-Tuning | Vector Embeddings & Semantic Search (FAISS) | PySpark & Databricks | NLP & Chatbot Development | Churn Modelling | Team Leadership | Python & SQL & PyTorch & TensorFlow | GDPR & HIPAA Compliant AI Deployment

KEY ACHIEVEMENTS

- Revenue Growth — Boosted Paytm's credit card product revenue by 40% within 3 months of deploying an optimised credit risk Probability of Default (PD) model for lending decisions.
- Chatbot Accuracy — Achieved 94% response accuracy on Paytm's Ask AI chatbot by selecting, fine-tuning, and deploying an LLM on domain-specific customer support data.
- Latency Reduction — Reduced LLM inference latency by 60% through async parallel batching and Azure OpenAI prompt caching across 2,600+ production batches on the DRL Pharma platform.
- Sales AI Scale — Scored and ranked 13,000+ customer-product opportunities with AI-generated priority scores, 9-driver business analysis, and ranked sales rep assignments.
- Fraud Detection Precision — Improved credit card fraud detection precision to 92% using XGBoost ensemble model with SMOTE-based class imbalance correction on real-world transaction data.
- Credit Risk Reduction — Reduced portfolio credit risk by 30% by enhancing a PD model assessing default probability for credit card applicants.

PROFESSIONAL EXPERIENCE

Senior Data Scientist | Navikenz | Noida, India *10/2025 – Present*

- Led a team of 5 Data Scientists to deliver end-to-end Generative AI and machine learning solutions for BFSI and pharmaceutical enterprise clients; managed sprint planning, code reviews, and executive stakeholder presentations.
- Architected the DRL Pharma Sales Intelligence platform — a multi-source RAG pipeline integrating sales, CRM interaction, opportunity, and product master data to generate AI-driven sales rep assignments, priority scores, and business driver analysis for 13,000+ customer-product combinations.
- Built a dual-embedding architecture using Azure OpenAI text-embedding-3-large (3072-dim vectors) and FAISS IndexFlatL2, combining deterministic exact-match retrieval with semantic similarity scoring for high-precision LLM context assembly.
- Implemented a two-phase hybrid retrieval strategy: Phase 1 guarantees exact customer-product matches; Phase 2 blends therapy alignment, recency weighting, and embedding similarity scores to fill remaining context slots.
- Engineered Format-Preserving Encryption (FPE/AES-FF3-1) and AES-256-GCM for PII field protection, ensuring full GDPR and HIPAA compliance while maintaining FAISS matching accuracy and LLM reasoning quality.
- Designed async parallel LLM processing with dynamic customer-aware batching and Azure OpenAI prompt caching, reducing inference latency by 60% and cutting token costs across 2,600+ production batches.

- Deployed scalable credit risk, delinquency prediction, and recovery optimisation models for BFSI clients using supervised and unsupervised machine learning on PySpark and Databricks.
- Built LLM-powered personalisation pipelines using vector embeddings, prompt engineering, and inference optimisation; partnered with product and engineering teams to productionise models end-to-end.

Senior Data Scientist | Vervent | Remote *03/2025 – 09/2025*

- Developed and enhanced AI-driven credit risk and collections models to identify delinquent and charge-off customers using supervised and unsupervised machine learning including Logistic Regression, Random Forest, and XGBoost.
- Performed customer risk segmentation using clustering algorithms (K-Means, DBSCAN) to enable data-driven recovery strategies and targeted offer optimisation.
- Applied Large Language Models (LLMs) and prompt engineering to generate personalised, empathetic debt recovery messaging at scale, improving customer engagement and response rates.
- Leveraged vector embeddings and semantic similarity search to align customer context with optimal offer templates and communication strategies.
- Validated model stability using KS statistic, AUC-ROC, Gini coefficient, and Population Stability Index (PSI) across deployment time windows to ensure production reliability.

Assistant Manager – Data Science | Paytm | Noida, India *07/2022 – 03/2025*

- Led development of the Ask AI LLM-powered in-app chatbot; selected, fine-tuned, and deployed a large language model on Paytm-specific domain data, achieving 94% response accuracy in production.
- Integrated the Ask AI chatbot with Paytm's backend infrastructure for real-time query resolution; implemented continuous learning loops from user interaction feedback for iterative model improvement.
- Expanded a credit risk model to assess Probability of Default (PD) for credit card applicants, directly driving 40% revenue growth within three months by optimising lending decisions.
- Built a propensity model to predict customer likelihood of credit card application, boosting conversion by aligning marketing campaigns with high-propensity customer segments.
- Enhanced NLP capabilities including intent recognition and context-aware response generation, reducing escalation rate to human agents and improving customer satisfaction scores.

Data Scientist | Metacube Software Pvt. Ltd. | Jaipur, India *11/2021 – 07/2022*

- Engineered a machine learning model for real-time credit card fraud detection using transaction patterns and time-series behavioural features; achieved 92% precision using XGBoost with SMOTE for class imbalance correction.
- Optimised model hyperparameters using Grid Search CV; evaluated performance using Precision, Recall, F1-Score, and AUC-ROC on held-out test sets.
- Built a customer churn prediction model to identify accounts at risk of closure; delivered actionable retention strategies based on model-driven customer segmentation.

Junior Software Engineer | .Kreate | Noida, India *05/2019 – 11/2021*

- Conducted end-to-end data preprocessing including feature engineering, outlier detection, normalisation, and data validation to support machine learning model development workflows.
- Built foundational expertise in Python, SQL, and machine learning frameworks through hands-on model development and data validation support, anchoring subsequent data science career growth.

KEY PROJECTS

DRL Pharma Sales Intelligence Platform | Navikenz *10/2025 – Present*

End-to-end GenAI-powered sales prioritisation system for Dr. Reddy's Laboratories using RAG, FAISS, and Azure OpenAI GPT-4o.

- Led 5-member data science team; owned solution architecture, sprint planning, code review, and full client delivery lifecycle.
- Designed multi-source data pipeline merging sales, CRM, opportunity, and product master data across 14,820+ unique customer-product combinations for LLM context enrichment.
- Built dual-embedding retrieval: text-embedding-3-large (3072-dim) vectors indexed in FAISS; two-phase hybrid retrieval combines exact match with therapy alignment, recency weighting, and semantic scoring.

- Implemented FPE/AES-FF3-1 for PII field-level encryption and AES-256-GCM for free-text at rest; maintained FAISS accuracy and LLM output quality with zero downstream impact.
- Delivered ranked sales rep assignments, 9-driver business analysis, and 5-decimal priority scores for 13,000+ opportunities exported to multi-sheet Excel workbooks with per-rep priority ranking.
- Tech stack: Python, Azure OpenAI GPT-4o, text-embedding-3-large, FAISS, PySpark, Pandas, AsyncIO, python-cryptography, openpyxl.

First Payment Default (FPD) Prediction Model | Vervent *03/2025 – 09/2025*

Machine learning system to identify high-risk credit card customers and reduce early delinquency at onboarding.

- Built FPD model using Logistic Regression, Random Forest, and XGBoost with feature engineering on transaction history and credit bureau data; validated using KS statistic, AUC-ROC, and PSI.

Ask AI – LLM-Powered Customer Support Chatbot | Paytm *01/2024 – 03/2023*

In-app LLM chatbot for Paytm customer support.

- Selected, fine-tuned, and deployed an LLM on Paytm domain-specific data; integrated with backend APIs for real-time resolution; built continuous learning loops from user feedback.

Credit Risk Probability of Default (PD) Model | Paytm *01/2022 – 11/2024*

Predictive model estimating default risk for credit card applicants; drove 40% revenue growth post-deployment.

- Applied Logistic Regression and Random Forest; assessed performance using ROC-AUC, Precision-Recall, and F1-Score; performed data cleansing, outlier removal, and feature scaling.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C

Machine Learning / Deep Learning: Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras, Random Forest, Logistic Regression, Isolation Forest

Generative AI / LLM: Azure OpenAI (GPT-4o, text-embedding-3-large), LangChain, LangGraph, Prompt Engineering, RAG, FAISS, Vector Embeddings, LLM Fine-Tuning

Big Data / MLOps: PySpark, Databricks, MLflow, Apache Spark

Data Engineering: Pandas, NumPy, openpyxl, asyncio, Feature Engineering

Encryption / Security: AES-256-GCM, Format-Preserving Encryption (FPE/AES-FF3-1), python-cryptography, GDPR, HIPAA

Cloud / Platforms: Microsoft Azure, Azure Machine Learning, Azure OpenAI Studio, Jupyter Notebook, VS Code

Data Visualisation: Matplotlib, Seaborn, Power BI

Model Evaluation Metrics: AUC-ROC, KS Statistic, PSI, Gini Coefficient, F1-Score, Precision-Recall, SHAP, Confusion Matrix

EDUCATION

Bachelor of Technology – Computer Science *2015 – 2019*

ABES Engineering College | Dr. A.P.J. Abdul Kalam Technical University (AKTU), Uttar Pradesh

Senior Secondary (Class XII) & Secondary (Class X) *2013 – 2015*

Delhi Public School